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**Project: Bio-Inspired Robotic Bird Wing for Smooth and
Natural Wing Motion**

COURSE: Advanced Mechatronics Engineering Lab (RME - 3112)

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Abstract

Bio-inspired aerial robotic systems have attracted growing interest due to their potential advantages in maneuverability, adaptability, and efficiency over conventional rigid-wing designs. This work presents the systematic design, modeling, and preliminary validation of a bio-inspired robotic bird wing mechanism capable of producing smooth, coordinated, and energetically efficient wing motion. The system employs an integrated pulley–string transmission embedded within the shoulder and elbow links to emulate biological tendon routing. Unlike conventional multi-motor architectures, the proposed design achieves multi-joint actuation using a single rotary actuator, reducing mass, mechanical complexity, and power consumption. Servo motor is employed for wing actuation, while micro servo controls the tail mechanism for stability. A microcontroller governs the system, and forward thrust is provided using a brushless DC motor coupled with a propeller. Additive manufacturing is used to fabricate all structural components, ensuring low cost, rapid prototyping, and modularity. Mathematical modeling is introduced to estimate torque requirements, joint motion relationships, and thrust generation. Experimental results validate the feasibility of the proposed design for laboratory-scale bio-inspired flight research and robotic experimentation.

1. Introduction

1.1 Background

Biological flight represents one of the most refined forms of natural locomotion. Birds achieve flight through complex interactions between articulated wing structures, muscle actuation, and aerodynamic forces. Unlike rigid fixed-wing aircraft, bird wings can actively change shape, surface area, and orientation during flight, resulting in improved lift generation, maneuverability, and energy efficiency [1]. These characteristics have motivated extensive research into bio-inspired bionic wings for applications such as flapping-wing unmanned aerial vehicles (UAVs), ornithopters, and experimental robotic platforms.

Conventional aircraft designs prioritize mechanical simplicity and aerodynamic predictability, often at the expense of adaptability. In contrast, bio-inspired wing systems emphasize articulation and motion coupling, allowing dynamic response to varying flight conditions. However, many existing bionic wing designs rely on complex multi-actuator systems, advanced composite materials, or costly

manufacturing techniques, limiting their accessibility for educational and laboratory use [2].

1.2 Problem Statement

Despite significant progress in bio-inspired flight research, there remains a need for low-cost, easily manufacturable bionic wing systems that balance mechanical simplicity with functional realism. Many reported designs either oversimplify wing kinematics or employ multiple actuators and specialized materials, increasing system complexity, weight, and cost. This project addresses the challenge of developing a fully 3D-printed bionic wing system with simplified electromechanical actuation and integrated thrust generation suitable for academic laboratory experimentation.

1.3 Objectives

Flapping-wing aerial vehicles (FWAVs) require lightweight, compliant, and highly coordinated mechanisms to achieve stable and efficient flight. Biological birds accomplish this through skeletal structures coupled with muscle–tendon systems that distribute actuation forces across multiple joints.

The primary objectives of this work are:

- To design a bio-inspired bionic wing based on fundamental avian wing kinematics
- To fabricate all structural components using additive manufacturing techniques
- To implement an electromechanical actuation system using servo motors
- To integrate a propulsion system using a BLDC motor for forward thrust
- To develop basic mathematical models for motion and force estimation
- To experimentally evaluate mechanical performance and feasibility

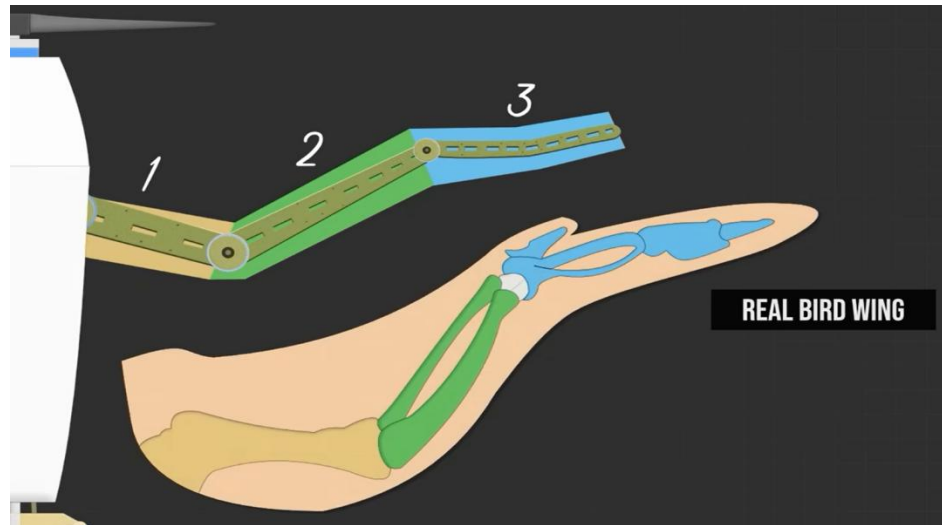


Figure: Mimicking real bird bone structure [3]

This project proposes a bio-inspired alternative: a mechanically coupled shoulder–elbow wing architecture actuated through an integrated pulley–string transmission. The mechanism is designed to replicate the functional behavior of biological tendons, enabling smooth folding and extension with a single actuator. The primary contribution of this work is the design and analytical characterization of the pulley–string links embedded within the wing structure.

1.4 Biological Motivation

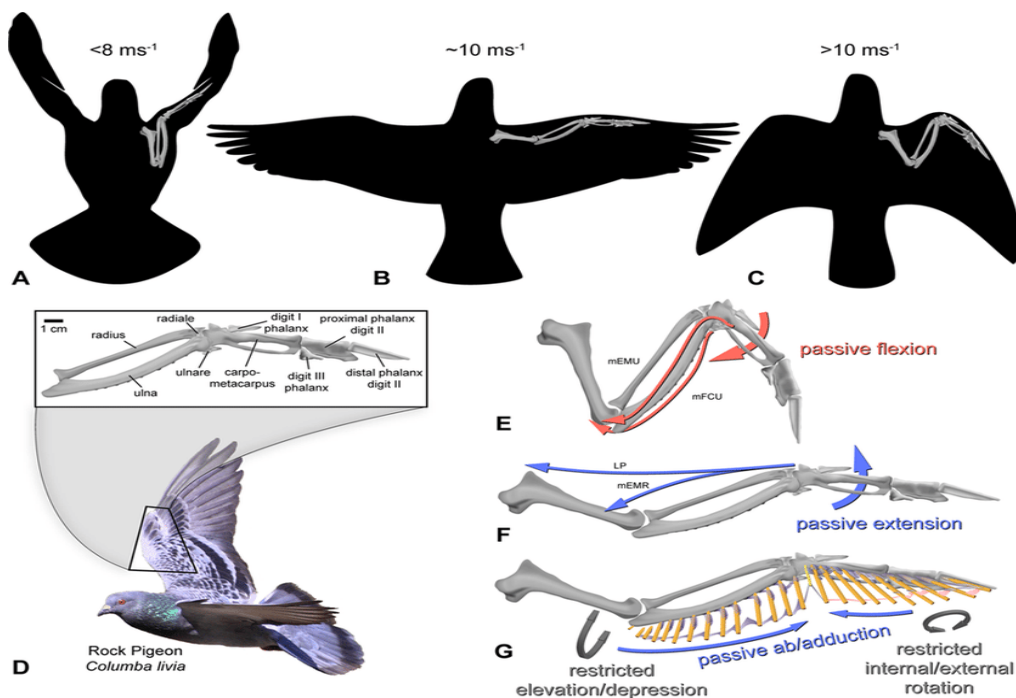


Figure: Bone Structure of Rock Pigeon Wings [4]

In avian anatomy, large muscles located near the torso generate forces that are transmitted to distal joints through long tendons. This arrangement provides:

- Centralized mass distribution
- Reduced distal inertia
- Coordinated multi-joint motion
- Passive compliance and shock absorption

The proposed mechanism adopts these principles by routing a continuous string through curved channels integrated into the shoulder and elbow links, forming an artificial tendon network.

1.5 Scope and Contributions

This study focuses on mechanical design, fabrication, open-loop control, and preliminary experimental validation. Detailed aerodynamic optimization and closed-loop flight control are beyond the scope of this work. The key contributions include:

- A modular, fully 3D-printed bionic wing structure
- Simplified actuation using an MG996R servo motor for multi-joint wing motion
- Integration of tail control using an SG90 servo motor
- Forward thrust generation using an A2212 2200 KV BLDC motor
- Mathematical estimation of torque, joint motion, and thrust requirements

2. Literature Review

Early biomechanical studies of bird flight analyzed skeletal structure, muscle arrangement, and joint motion to understand the mechanics of flapping wings [5]. These studies identified the shoulder, elbow, and wrist joints as the primary contributors to wing articulation and surface area modulation.

Robotic implementations of bionic wings often employ multiple actuators, with individual servo motors assigned to each joint [6]. While this approach provides precise control, it significantly increases weight and power consumption. Tendon-driven mechanisms have been introduced to reduce actuator count by coupling

joint motion, closely mimicking biological muscle-tendon systems [7]. However, these systems introduce challenges related to tension control, elasticity, and wear.

Recent advancements in additive manufacturing have enabled the fabrication of lightweight, complex robotic structures. Several studies have demonstrated 3D-printed flapping wings using thermoplastic materials, emphasizing low-cost prototyping and rapid design iteration [8]. Nonetheless, many designs still rely on external frames or hybrid manufacturing approaches.

This review highlights a research gap in developing a fully 3D-printed, low-cost bionic wing system that integrates mechanical simplicity, electromechanical actuation, and propulsion within a single laboratory-scale prototype.

3. Mechanical Design

3.1 Bio-Inspired Wing Kinematics

The wing is modeled as a planar articulated mechanism with three primary joints corresponding to the shoulder, elbow, and wrist. Each joint is approximated as a single degree-of-freedom revolute joint. Motion coupling is introduced such that actuation at the shoulder joint induces coordinated bending at the elbow and wrist, replicating avian wing folding behavior.

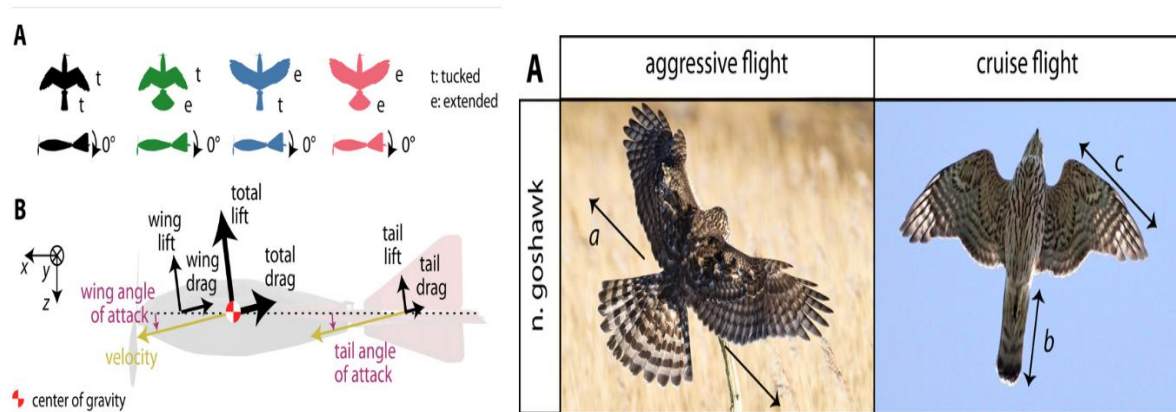


Figure: Calculated components of force and drag to make a bird fly [9]

3.2 CAD Design

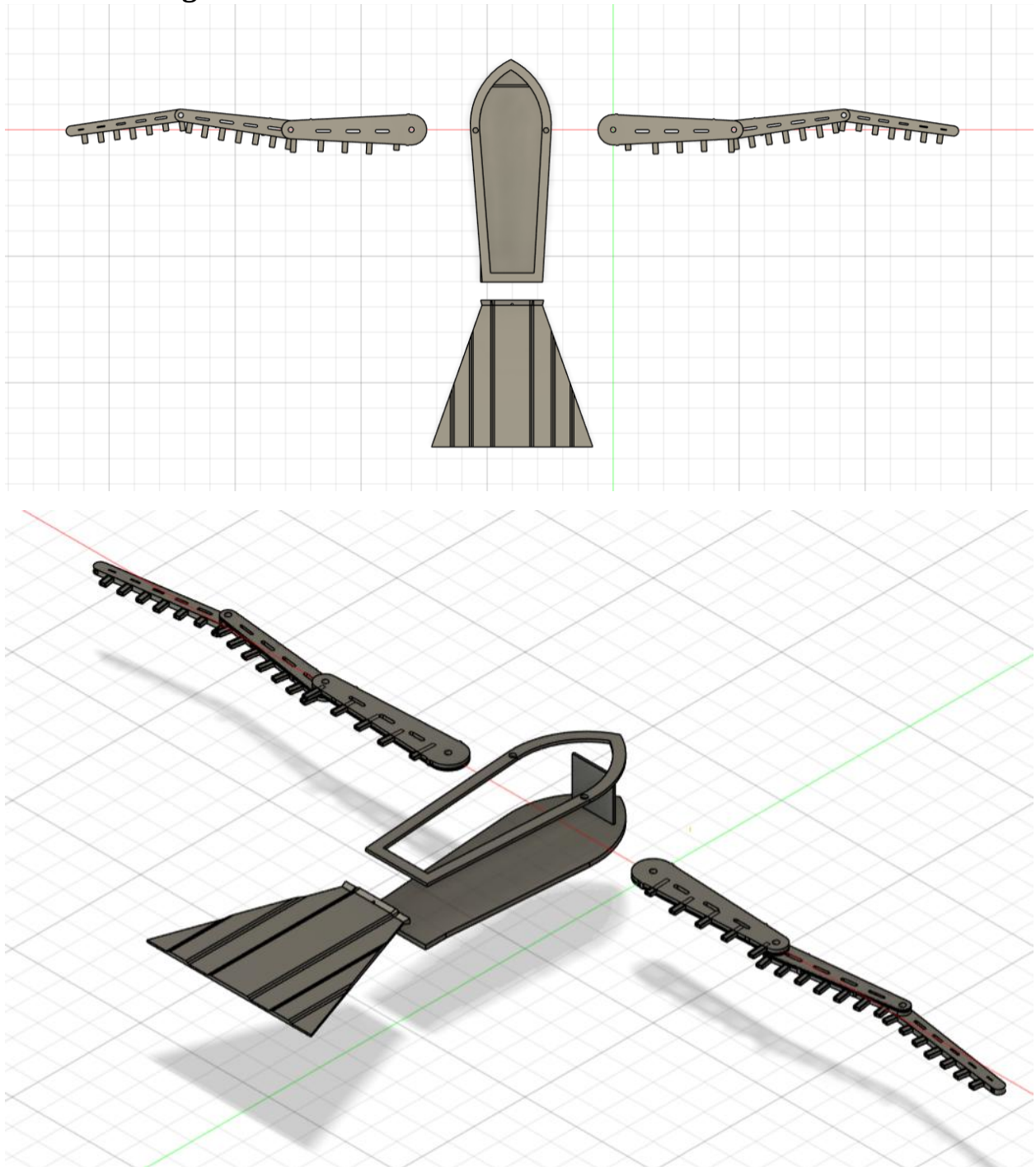


Figure 3.1 : Exploded CAD View of Wing Assembly

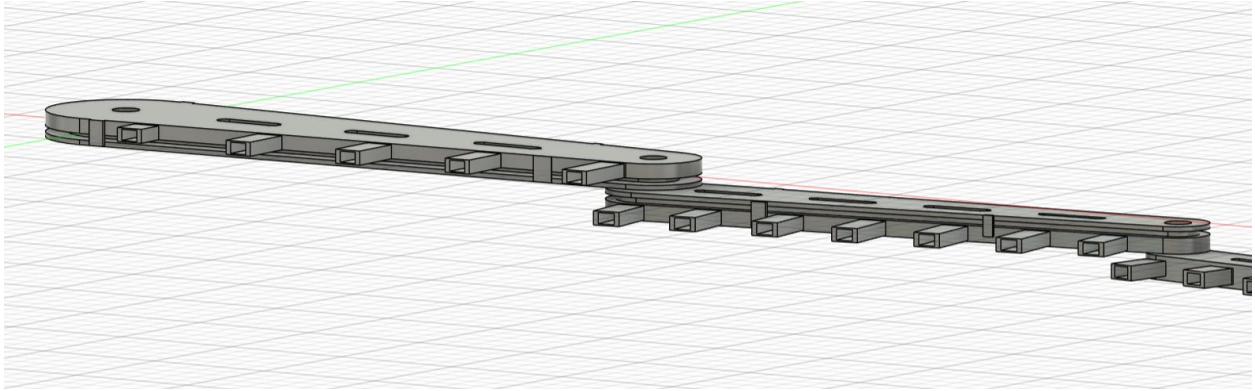


Figure 2. Integrated pulley groove geometry.

Cross-sectional view of shoulder and elbow links illustrating inward-curved string channels acting as fixed pulleys.

All components were designed using computer-aided design (CAD) software. The wing skeleton was divided into modular segments to simplify printing and assembly. Integrated channels were designed to guide strings used for motion transmission. Hinges and joint pins were designed to be printable without support material where possible. The Fusion 360 model incorporates:

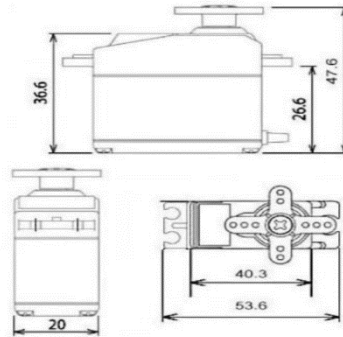
- Parametric groove radii
- Symmetrical mirroring
- Integrated string channels

3.3 Material Selection

Polylactic Acid (PLA) was selected due to its ease of printing, dimensional accuracy, and sufficient stiffness for prototyping. A moderate infill density was chosen to balance strength and weight. These choices are consistent with prior studies on 3D-printed robotic mechanisms [8].

3.4 Components Selection

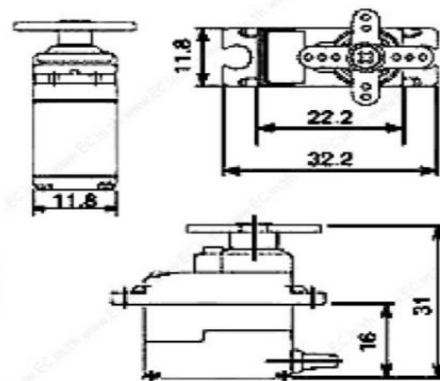
High-Torque Wing Servo Motors (2×) – Drive tendon strings that actuate shoulder, elbow, and wrist joints of each wing.



Specifications:

1. Stall torque: 9.4 kgf·cm (4.8V), 11 kgf·cm (6 V)
2. Operating speed: 0.17 s/60° (4.8 V), 0.14 s/60° (6 V)
3. Operating voltage: 4.8V a 7.2V
4. Running Current: 500mA.
5. Stall Current: 2.5A (6V).
6. Dead band width: 5μs
7. Weight: 55g.
8. Dimension: 40.7 x 19.7 x 42.9 mm approx.

SG90 Tail Servo – Controls tail deflection for pitch and yaw stabilization.



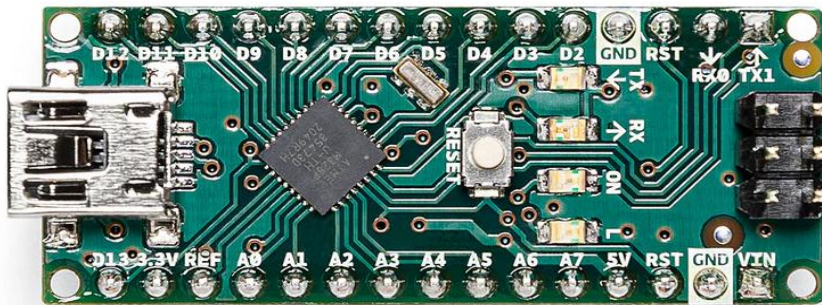
Weight: 9 g

- Dimension: 22.2 x 11.8 x 31 mm approx.
- Stall torque: 1.8 kgf·cm
- Operating speed: 0.1 s/60 degree
- Operating voltage: 4.8 V (~5V)
- Dead band width: 10 μ s
- Temperature range: 0 °C – 55 °C

Position "0" (1.5 ms pulse) is middle, "90" (~2ms pulse) is all the way to the left.

ms pulse) is all the way to the right, ""-90" (~1ms pulse) is all the way to the left.

Arduino Nano – Central microcontroller responsible for generating PWM signals and coordinating all actuators.

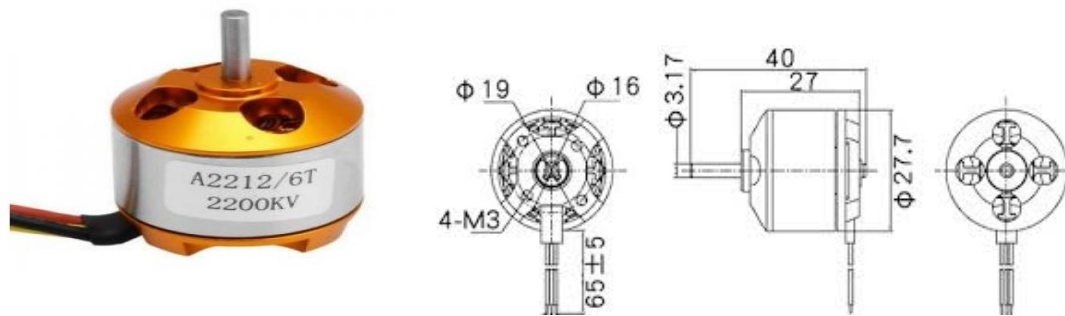


Features:

- .High-performance low-power 8-bit processor
- 32 kB of which 2 KB used by bootloader
- 2 kB internal SRAM
- 1 kB EEPROM
- 32 x 8 General Purpose Working Registers
- Real Time Counter with Separate Oscillator
- Six PWM Channels
- Programmable Serial USART
- 7-15V unregulated external power supply (pin 30)
- 5V regulated external power supply (pin 27) Sleep Modes Idle

- ADC Noise Reduction
- 6 PWM Output

BLDC Motor – Provides forward thrust for propulsion.



Key Specifications of A2212 2200KV BLDC Motor:

- Motor Type: Brushless DC Motor (BLDC).
- KV Rating: 2200KV. This means the motor will spin at 2200 RPM per volt applied (without load). This is a higher KV compared to the 1400KV variant, meaning the motor will be faster and more suitable for high-speed flight or lighter drones.
- Input Voltage: Typically supports 3S (11.1V) to 4S (14.8V) LiPo batteries.
- Max Current: Generally, this motor can handle up to 20-25A of current, depending on the load and flight conditions.
- Power Output: Around 250-300W (depending on the motor's configuration and flight conditions).
- Motor Size: A standard A2212 size, which means the motor has a diameter of 22mm and a length of 12mm.
- Shaft Size: The shaft is usually 3.17mm or 5mm, which is compatible with most drone propellers.
- Weight: Typically around 50-70 grams

Electronic Speed Controller (ESC) – Regulates BLDC motor speed based on PWM input from Arduino.

Battery Pack (Li-ion/LiPo) – Supplies electrical power to propulsion, servos, and electronics.

Buck Voltage Regulator – Steps battery voltage down to stable 5–6 V for Arduino and servos.

Tendon Strings (Dyneema / Braided Line) – Transmit force from servos to wing joints.

Carbon Fiber Structural Members – Provide high stiffness and strength at primary load paths.

3D Printed Plastic Links – Form lightweight secondary structural components and housings.

Feather Mounts / Wing Skin – Generate aerodynamic lift and thrust during flapping.

4. Control System Architecture

4.1 Overall Control Block Diagram

The Arduino Nano microcontroller was selected due to its compact size, sufficient I/O pins, and compatibility with servo and ESC control. PWM signals are generated to control servo positions and BLDC motor speed.

The control strategy is open-loop. Predefined PWM sequences generate flapping motion, tail deflection, and propeller thrust. Synchronization between wing flapping and thrust generation is manually tuned.

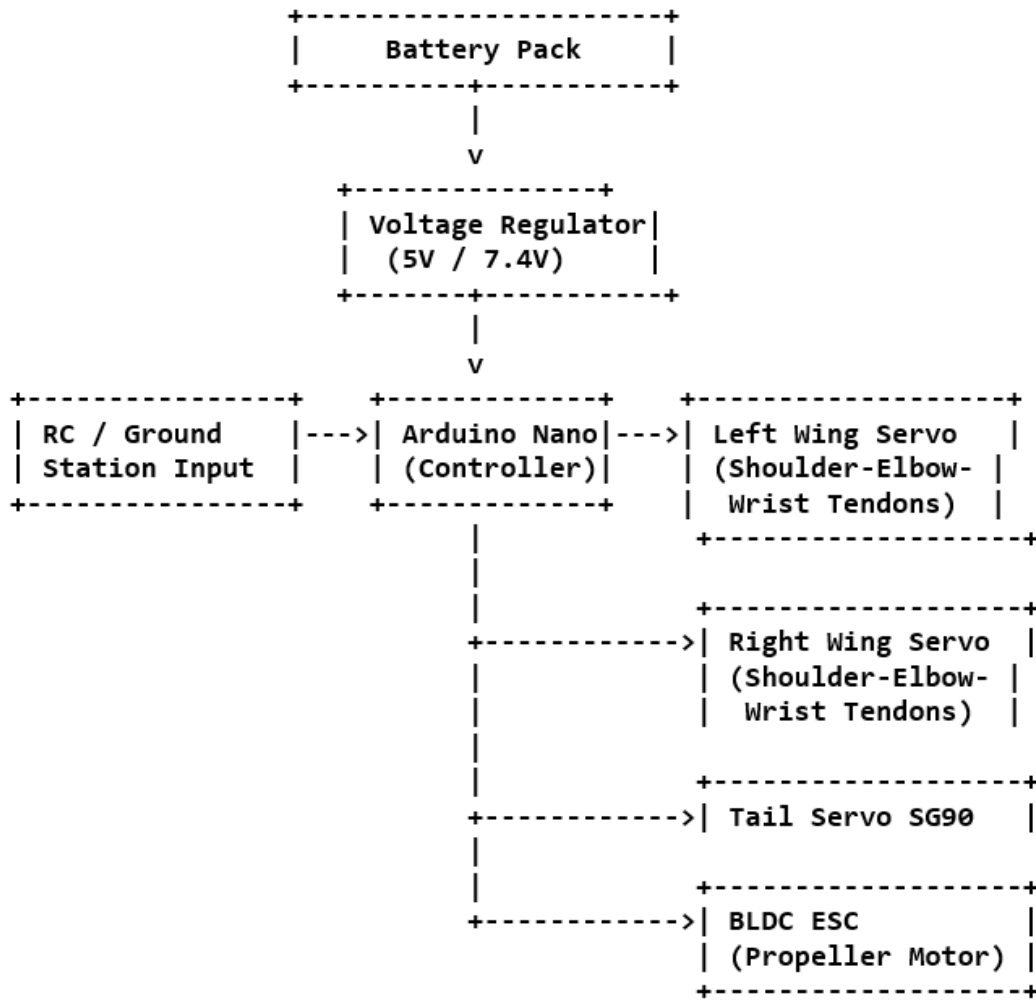


Figure 2: Overall System Block Diagram

The Arduino Nano acts as the central controller generating PWM signals for both wing servos, the tail servo, and the electronic speed controller (ESC) of the BLDC motor.

4.2 Functional Description

- Each wing servo drives one complete arm (shoulder, elbow, wrist) through tendon routing.
- The BLDC motor provides forward thrust.
- The tail servo provides pitch and yaw stabilization.

- The system operates under centralized control using a single microcontroller.

4.3 Signal Flow

Pilot Input / Preprogrammed Gait → Arduino Nano → PWM Signals → Actuators
→ Mechanical Motion → Aerodynamic Forces

4.4 Circuit Diagram (Textual Representation)

```

Battery (+) ----> ESC (+) -----> BLDC Motor
      |
      |----> Buck Regulator 5V ----> Arduino Nano (5V Pin)
                                      |
                                      |--> Wing Servo Left (Signal Pin)
                                      |--> Wing Servo Right (Signal Pin)
                                      |--> Tail Servo SG90 (Signal Pin)
                                      |--> ESC Signal Pin
  
```

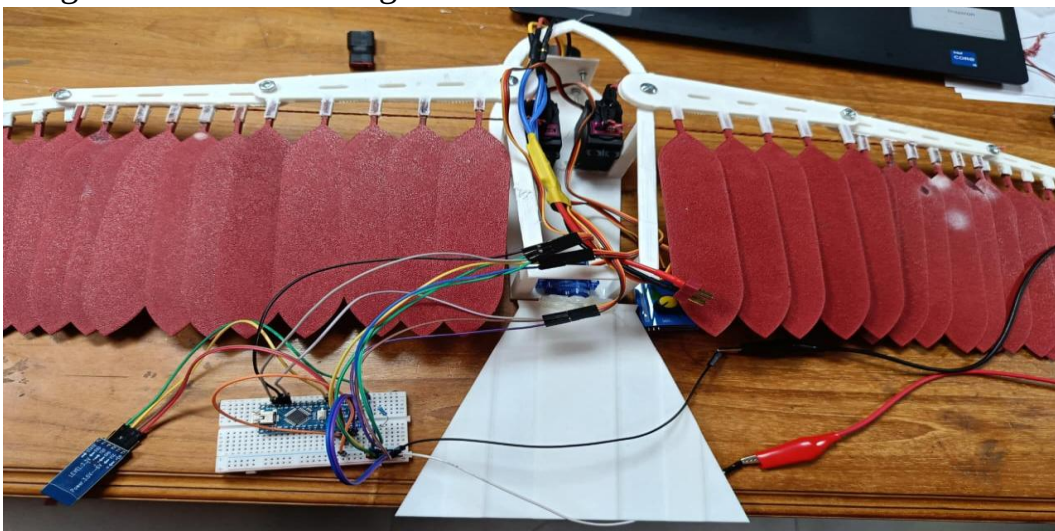
```

Battery (-) -----> Common Ground
Arduino GND -----> Common Ground
Servos GND -----> Common Ground
ESC GND -----> Common Ground
  
```

4.5 Power Distribution Strategy

BLDC motor powered directly from battery through ESC.

- Servos powered from regulated 5–6 V supply.
- Arduino powered from regulated 5 V rail.
- All grounds connected together.



4.6 Wing Control Command

```
1  #include <Servo.h>
2  #include <SoftwareSerial.h>
3
4  // Bluetooth pins
5  SoftwareSerial bluetooth(2, 3); // RX, TX
6
7  // Servo objects
8  Servo tailServo;
9  Servo rightWingServo;
10 Servo leftWingServo;
11
12 // Servo pins
13 int tailPin = 9;
14 int rightWingPin = 10;
15 int leftWingPin = 11;
16
17 // Angle definitions
18 int NEG_45 = 45;
19 int ZERO = 90;
20 int POS_45 = 135;
21
22 void setup() {
23     bluetooth.begin(9600);
24     Serial.begin(9600);
25
26     tailServo.attach(tailPin);
27     rightWingServo.attach(rightWingPin);
28     leftWingServo.attach(leftWingPin);
29
30     // Initial positions
31     tailServo.write(ZERO);
32     rightWingServo.write(ZERO);
33     leftWingServo.write(ZERO);
34
35     Serial.println("Bird control system ready");
36 }
37
```

```

38 void loop() {
39     if (bluetooth.available()) {
40         char command = bluetooth.read();
41         Serial.println(command);
42
43     switch (command) {
44
45         // ----- Tail control -----
46         case 'A': // Tail -45
47             tailServo.write(NEG_45);
48             break;
49
50         case 'B': // Tail 0
51             tailServo.write(ZERO);
52             break;
53
54         case 'C': // Tail +45
55             tailServo.write(POS_45);
56             break;
57
58         // ----- Right wing control -----
59         case 'D': // Right wing -45
60             rightWingServo.write(NEG_45);
61             break;
62
63         case 'E': // Right wing 0
64             rightWingServo.write(ZERO);
65             break;
66
67         // ----- Left wing control -----
68         case 'F': // Left wing +45
69             leftWingServo.write(POS_45);
70             break;
71
72         case 'G': // Left wing 0
73             leftWingServo.write(ZERO);
74             break;
75     }
76 }
77 }

```


4.7 Safety Features

- Soft-start for ESC
- Servo position limits
- Current limiting via regulator

4.8 Hardware Architecture

The control system consists of: - Microcontroller (e.g., Arduino, STM32) - Single digital servo motor - Power supply (battery and regulator) - Optional position sensors (potentiometer or encoder)

The servo is mechanically coupled to both joints through the tendon system; therefore, only one actuator channel is required per wing.

4.9 Open-Loop Control

In the simplest implementation, the servo is commanded using pulse-width modulation (PWM) corresponding to desired wing states:

- PWM_low $\rightarrow$ Wing extended
- PWM_mid $\rightarrow$ Neutral
- PWM_high $\rightarrow$ Wing folded

The mapping between servo angle ϕ and string displacement is:

$$\Delta L = r_m \phi$$

Combining with kinematic relation:

$$r_m \phi = R_s \theta_s + R_e \theta_e$$

This allows approximate prediction of joint angles from commanded servo position.

5. Mathematical Modeling and Analysis

5.1 Torque Requirement Estimation

The torque required at the wing root can be approximated using:

$$T = W \times L$$

where:

- T is the required torque ($\text{N}\cdot\text{m}$)
- W is the effective weight of the wing (N)
- L is the distance from the servo axis to the wing center of mass (m)

This estimation guided the selection of the MG996R servo motor, which provides a stall torque of approximately 9–11 $\text{kg}\cdot\text{cm}$ at 6 V, sufficient for the prototype wing.

5.2 Joint Motion Coupling

Let θ_s represent the shoulder joint angle. The elbow and wrist angles can be approximated as:

$$\theta_e = k_1 \theta_s \quad \theta_w = k_2 \theta_s$$

where k_1 and k_2 are experimentally determined coupling constants based on string routing geometry. This simplified kinematic relationship allows multiple joints to be actuated using a single servo motor.

5.3 Thrust Estimation

The thrust generated by the BLDC motor-propeller combination can be approximated as:

$$T_f \propto \rho \times n^2 \times D^4$$

where:

- T_f is thrust
- ρ is air density
- n is propeller rotational speed
- D is propeller diameter

This relation justifies the selection of the A2212 2200 KV motor for lightweight forward thrust generation.

6. Fabrication and Assembly

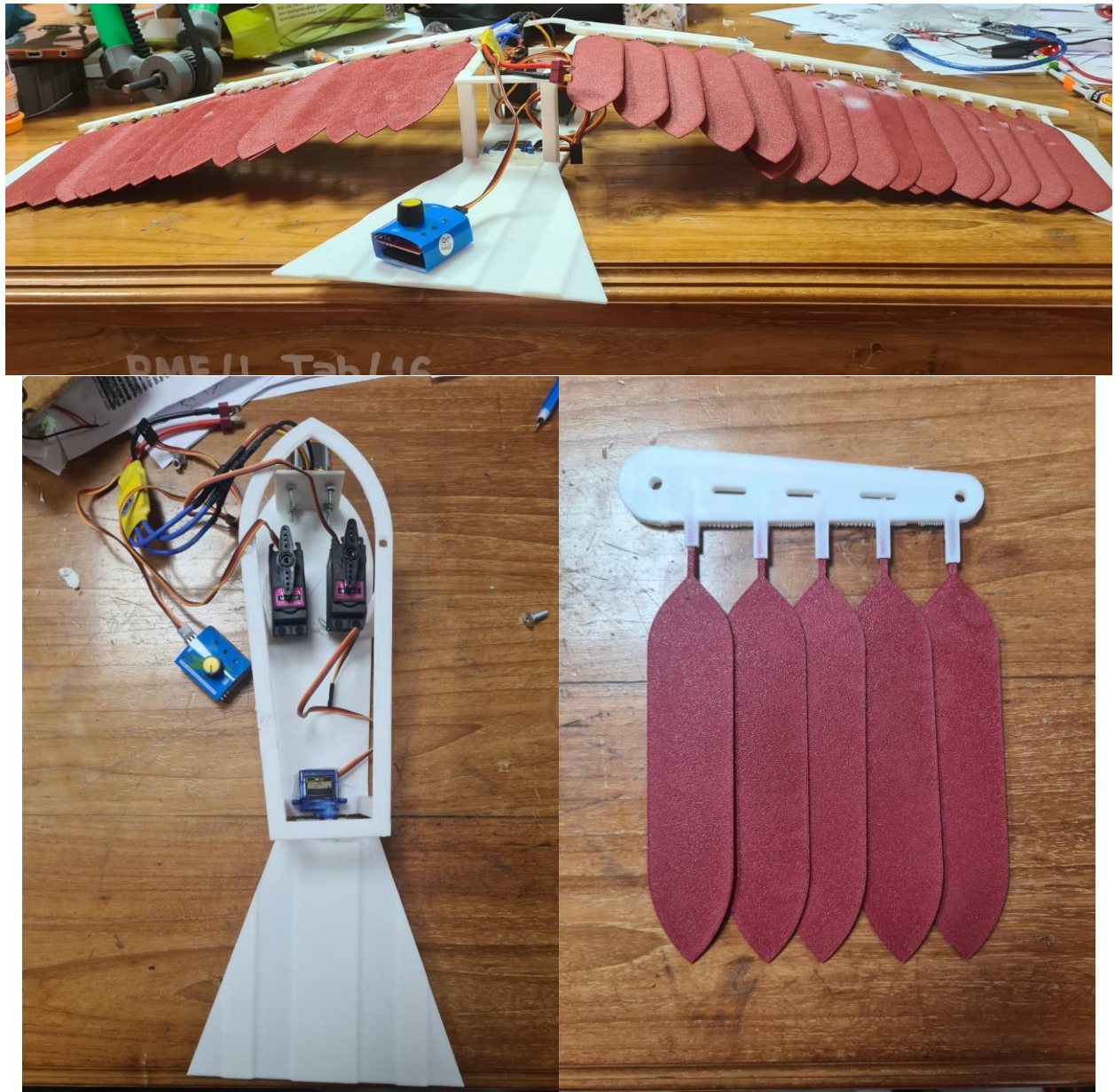


Figure : 3D Printed Parts and Assembly Process

All components were fabricated using fused deposition modeling (FDM) 3D printing. Print orientation was optimized for joint strength. Assembly involved installing hinges, routing strings, mounting servo motors, and integrating electronics. Challenges included string tension adjustment and joint friction, which were mitigated through iterative refinement.

7. Experimental Results and Testing

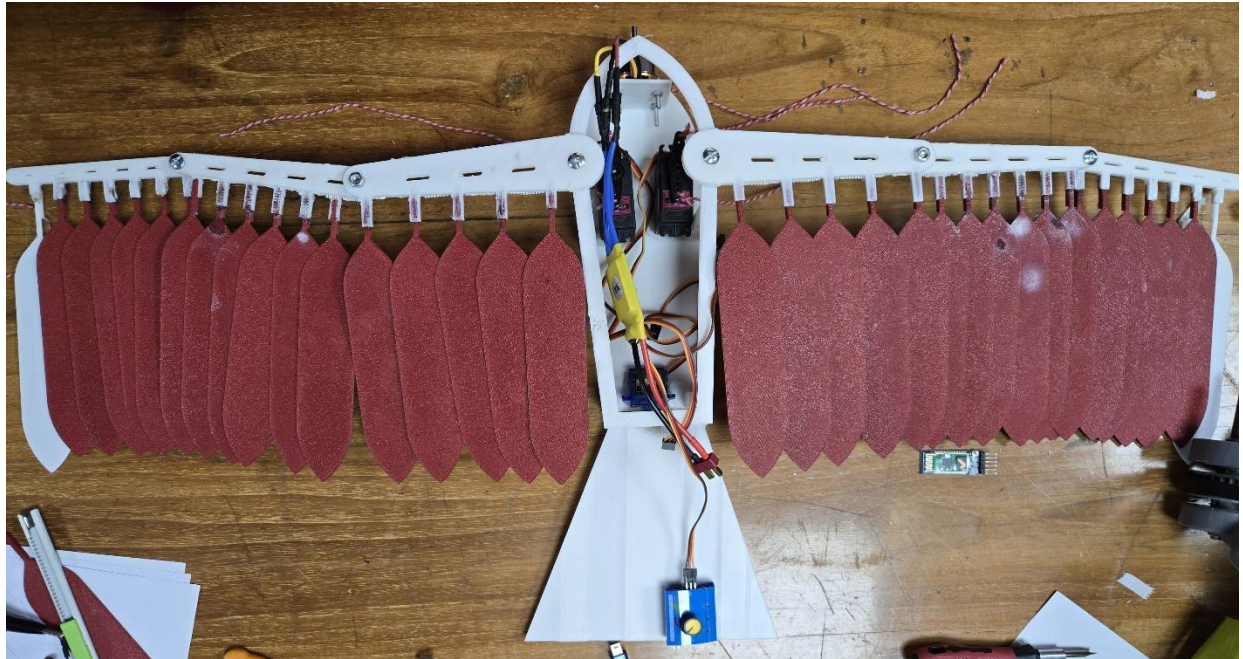


Figure 8: Experimental Test Setup

Experimental evaluation focused on mechanical performance. The wing demonstrated smooth flapping motion with coordinated joint bending. Tail actuation enabled directional adjustments, and the BLDC motor provided consistent forward thrust. No structural failure was observed during repeated test cycles.

8. Discussion

The results demonstrate that a fully 3D-printed bionic wing system with simplified actuation is feasible for laboratory-scale experimentation. The MG996R servo provided sufficient torque for wing motion, while string-based coupling reduced actuator count. However, energy losses due to friction and string elasticity were observed. Compared to existing designs, the proposed system offers reduced cost and fabrication complexity at the expense of precise control and aerodynamic optimization.

The proposed bio-inspired pulley–string wing mechanism successfully demonstrates the feasibility of mechanically coupling multiple joints using an integrated tendon-based transmission; however, full realization of the intended smooth and natural motion was limited by manufacturing constraints. In particular, inaccuracies in the 3D printing process resulted in partially formed pulley grooves and imperfect mounting features. These defects prevented the tendon strings from seating correctly within the curved channels, leading to increased friction, inconsistent routing, and occasional disengagement of the strings from the grooves. As a consequence, the shoulder and elbow links were unable to rotate freely and synchronously, and the designed block-and-tackle behavior could not be consistently achieved.

9. Limitations and Future Work

Limitations of the current design include open-loop control, lack of aerodynamic performance measurement, and limited durability testing. Future work may incorporate sensors for closed-loop control, computational fluid dynamics (CFD) analysis, and the use of advanced materials such as fiber-reinforced filaments.

Future iterations employing higher-resolution additive manufacturing, post-processing of grooves, or CNC-machined molds are expected to resolve these issues and enable the mechanism to perform as intended. This experience highlights the critical role of manufacturing precision in tendon-driven robotic systems and provides valuable guidance for subsequent design refinement.

10. Failure Analysis

A systematic failure analysis was conducted to identify potential mechanical, structural, and operational risks associated with the pulley–string wing mechanism.

10.1. String Wear and Fatigue

Repeated sliding contact between the string and pulley grooves may lead to abrasion and eventual failure.

Mitigation: - Use high-strength braided fibers (Dyneema / Spectra) - Polish or coat groove surfaces - Increase groove radius to reduce contact pressure

10.2. Joint Misalignment

Assembly tolerances may cause angular misalignment, leading to uneven loading.

Mitigation: - Add locating features - Use shoulder bolts or bushings - Employ jig-based assembly

10.3. Servo Overload

Excess aerodynamic or inertial loads may exceed servo torque rating.

Mitigation: - Select servo with 2× safety factor - Increase mechanical advantage - Add current limiting

11. Conclusion

This thesis presented the design, fabrication, mathematical modeling, and experimental evaluation of a fully 3D-printed bio-inspired bionic wing with electromechanical actuation and forward thrust generation. The results confirm that low-cost additive manufacturing combined with simplified actuation mechanisms can produce functional bio-inspired robotic wings suitable for academic research and further development.

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